

today are:

1. ARINC 429 (A429)
2. MIL-STD-1553B, also covered by Def Stan 00-18 Part 2 and STANAG 3839

3. ARINC 629 (A629)
4. Fibre-optic bus

Of these, A429 and A629 are commonly in use on civilian aircrafts. MIL-STD-1553 is a military standard.

Integrated flight management system

Today, the glass cockpit uses several displays driven by integrated flight management systems that can be adjusted to display flight information as needed.

Auto-pilots integrate both fly-by-wire of flight controls and power-by-wire engine controls. These generally divide a flight into taxi, take-off, ascent, level, descent, approach, landing and taxi phases. Landing on runway and controlling the aircraft on rollout, i.e., keeping it at the centre of the runway, is CAT-3B landing, used on the majority of major runways today. Landing, rollout and taxi control to stand is CAT-3C.

Basically, integrated flight control systems should control the primary and secondary flight control surfaces. Primary control surfaces are pitch, yaw and roll motions of the aircraft. Secondary flight control surfaces help the aircraft to take off and land like leading-edge slats and spoilers. The horizontal stabiliser at the rear controls the pitch motion, the rudder yaw or direction, while the Ailerons control the roll motion of the aircraft.

On-board computers with associated software control the flight of the aircraft. Both primary and secondary flight control surfaces are automatically controlled, reducing the workload of the pilot.

The autopilot controls the engine power and is familiarly called power-by-wire concept. Full authority digital engine control (FADEC) enables matching of air-fuel ratio at various speeds, loads and altitudes to produce optimal power and reduce emissions.



Airbus A380 cockpit (Courtesy: Wikipedia)

FADEC is integrated to the autopilot.

GPS devices in cockpits

GPS provides geographical location in term of latitude and longitude of the destination.

Pilots can use the touchscreen panel to check information about nearest airports, weather reports, runway patterns and distance. When a plane lands, a button push gives the pilot the more familiar car GPS screens. GPS systems used in aircrafts now have an accuracy of 1 cm. This is accomplished by using GPS in conjunction with accurate altimeters and differential GPS. The differential GPS interacts with airports (base stations) in addition to satellites.

Some manufacturers provide landing facility with GPS alone or with an option for ILS (Instrument Landing System) facility. Most business jets use GPS for landing.

Basic flight instruments

Various manufacturers implement different features and philosophies in the cockpit flight management system. All of them aim at reducing the pilot's workload and situational awareness from take-off to landing.

The primary component of glass cockpits is the electronic flight instrument system, which displays all the information about the aircraft's situa-

tion, position and progress. Comprising left- and right-side primary flight displays and multi-function display screens, it primarily covers horizontal and vertical position, but also indicates time and speed. The second part of the glass cockpit, comprising over- and under-centre display screens, shows the aircraft's system conditions and engine performance.

Majority of the system-related controls (such as electrical, fuel, hydraulics and pressurisation), for example, are usually located in the ceiling on an overhead panel. Radios are controls such as the autopilot, usually placed just below the windscreen and above the main instrument panel on the glareshield.

As aircraft displays have advanced, so have the sensors that feed them.

Traditional gyroscopic flight instruments have been replaced with altitude and heading reference systems and air data computers, improving reliability and reducing cost and maintenance. GPS receivers are integrated into glass cockpits.

EICAS/ECAM warning systems

While on-board computers manage all the systems and take full control of the aircraft flight and propulsion systems, central advisory warning system draws